MARIANA MACEDO SANTOS

Computer Science • Focus on Data Science

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Summary

Computer Science student (UFOP, graduating 12/2026) with a special interest in data science. At TerraLab, I build ETL pipelines in Python and SQL. I have machine learning, data analysis and back-end development projects, and did an exchange in Industrial Data Science in Austria. Looking for a junior role in tech.

Education

B.Sc. in Computer Science **2022 – 12/2026**
Federal University of Ouro Preto (UFOP), Brazil

Exchange Student – Industrial Data Science **02/2026 – 07/2026**
Montanuniversität Leoben, Austria

- Hands-on specialization in Industrial Data Science, applied Machine Learning and Process Optimization.

IT Technician (technical high school degree) **2019 – 2021**
Federal Institute of Northern Minas Gerais (IFNMG), Brazil

Experience

Data Analytics – TerraLab, UFOP **08/2026 – Present**

- Develop and optimize ETL pipelines in Python and SQL to load and process large volumes of data.
- Manage and monitor environments with Apache Airflow (orchestration), Apache Doris (data warehouse) and Apache Superset (dashboards).

Undergraduate Research Assistant – Data Automation and Pipelines **UFOP, 2023**

- Developed Python scripts to automate the extraction, cleaning and integration of data.

Vice-President – Computer Science Student Association **UFOP, 2024 – 2026**

- Organized academic events and technical workshops, managed student teams and represented students.

Projects

Brazilian Economic Indicators Pipeline *Airflow, PostgreSQL, SQL, Superset, Docker*

- Daily Airflow pipeline that extracts five indicators from the Central Bank of Brazil API, organizes the data in layers (raw, staging and analytics) in a PostgreSQL data warehouse and feeds a Superset dashboard, with incremental, idempotent loads.

Undergraduate Thesis – Automatic Classification of Programming Error Messages *Python, scikit-learn, PyTorch*

- Comparison of models that classify C compilation errors by programming topic, as a basis for educational recommendation: Logistic Regression, Random Forest, SVM, an MLP neural network and the UniXcoder code model (embeddings with SVM, and fine-tuning).

Video Transcription App *Node.js, Express, PostgreSQL, Firebase, Next.js*

- REST API in Node.js/Express with a layered architecture, Firebase authentication, PostgreSQL persistence, asynchronous file processing and per-user usage quotas.

Transphobic Hate Speech Detection with BERTimbau *Python, PyTorch, Transformers*

- Collected tweets via the X API and fine-tuned a Portuguese BERT model to classify them (pair project).

Technical Skills

- **Programming Languages:** Python, SQL, R, JavaScript/TypeScript, Java, C/C++.
- **Data & Pipelines:** SQL, PostgreSQL, Relational Modeling, ETL, Apache Airflow, Apache Doris.
- **Analysis & Visualization:** Pandas, NumPy, R, statistical analysis, Excel, Power BI, Apache Superset, Matplotlib.
- **Machine Learning & NLP:** scikit-learn, PyTorch, Hugging Face Transformers (BERT, UniXcoder), TF-IDF.
- **Software Engineering:** Git/GitHub, Docker, REST APIs, Python (Flask), Node.js (Express).
- **Courses:** Pandas for Data Analysis (Kaggle), Advanced Python (Udemy), REST APIs with Flask and Python (Udemy).
- **Languages:** Portuguese (native), English (advanced).